

ANANNYA PRIYADARSHINI NEOG

San Jose, CA, USA | aneog@ucsc.edu | anannya903@gmail.com | +1 (831)-529-7340
[LinkedIn](#) | [Github](#) | [portfolio](#)

SUMMARY

Graduate student in Computer Science and Engineering with experience in **AI/ML, NLP, and full-stack development**, focused on building **AI-powered applications with intuitive user interfaces**. At UCSC, conducting research on **hybrid Retrieval-Augmented Generation (RAG) systems with Knowledge Graphs**, while serving as a Teaching Assistant for Data Structures and AI-focused coursework. Developed end-to-end systems including **AI-driven legal QA platforms, audio processing pipelines, and sign-language translation applications**, integrating **LLM-based retrieval, vector search, and interactive React-based interfaces**.

Skilled in **Python, JavaScript/TypeScript, React, and modern ML frameworks**, with experience in **vector databases, graph databases, and API-driven architectures**. Interested in building **scalable AI-backed systems with strong emphasis on user experience and product design**.

EDUCATION

University of California, Santa Cruz **Sep 2024 – Present**

Master of Science (MS) in Computer Science and Engineering GPA: 3.76/4

- **Relevant Coursework:** Artificial Intelligence, Neural Computation, Analysis of Algorithms, Interactive Storytelling (for Game Development), Machine Learning, Foundations of Deep Learning, Computer Networks, Machine Learning in Natural Language Processing

SRM Institute of Science and Technology, Kattankulathur, Chennai **July 2019 – May 2023**

Bachelor of Technology (B.Tech.) in Computer Science and Engineering GPA: 3.98/4

- **Relevant Coursework:** Artificial Intelligence, Natural Language Processing, Machine Learning, Software Engineering and Project Management, Data Structures and Algorithms, Database Management Systems, Data Mining and Analytics, Digital Image Processing, Genetic Programming

WORK EXPERIENCE

Graduate Researcher **Sept 2025 – Present**

AI Explainability and Accountability (AIEA) Lab, UCSC *Santa Cruz, California*

- Built an AI-powered legal QA system using **Python, Sentence-Transformers, and Chroma**, enabling semantic retrieval over legal documents and generating context-aware natural language responses.
- Implemented hybrid retrieval by combining **top-k vector search with knowledge graph expansion (Neo4j)**, where retrieved document chunks were enriched using graph neighbors (shared sections, entities), then merged and deduplicated to improve contextual relevance.
- Designed and developed a chat-based web application using **React.js, TypeScript, and REST APIs**, enabling conversational querying with real-time responses, structured answer rendering, and source-cited outputs.
- Enhanced user experience by implementing **chat history, retrieval mode switching, and dynamic filtering controls**, enabling users to explore and compare AI-generated results through an intuitive and interactive interface.

Graduate Teaching Assistant **Jan 2025 – Present**

University of California Santa Cruz *Santa Cruz, California*

- **CSE 101: Introduction to Data Structures and Algorithms**
- **MUSC 80L: Artificial Intelligence and Music**
 - * Assisted students with understanding DSA and working on algorithmic lab assignments, applying Python and neural network techniques for tasks such as musical note generation and audio synthesis, providing guidance on both conceptual understanding and hands-on implementation.
 - * Evaluated and provided constructive feedback on hundreds of student assignments and projects, ensuring fairness, accuracy, and Proctored and evaluated over hundreds of students' assignments and exams.

Big Data Analytics Intern **July 2021 – Sept 2021**

REGex Software Services *Rajasthan, India*

- Built data ingestion and transformation pipelines using **Apache Spark (PySpark), Talend ETL, and Apache Hive**, processing **10K RedBus booking records**; performed data cleaning (null handling, type casting, deduplication) and derived metrics such as booking trends, route demand, and revenue distribution using **Spark SQL**.
- Developed interactive dashboards using **Power BI** to visualize booking patterns, occupancy trends, and revenue insights; enabled intuitive data exploration by translating processed datasets into structured, user-friendly visual reports.

Web Development Intern **May 2020 – June 2020**

InMovidu Tech *Bengaluru, India*

- Built a responsive frontend for a movie ticket booking system using **React.js, JavaScript, HTML, CSS**, implementing seat selection, showtime browsing, and checkout flows; integrated with backend **microservices via REST APIs** to display real-time availability and pricing for a platform serving **50K users**.
- Developed reusable UI components using **React Hooks** and **Axios**, managing client-side booking workflows and handling API loading and error states across distributed services to deliver a smooth user experience.

RESEARCH STUDIES AND PROJECTS

Audio Genre Blending: Optimization-Based Audio Style Transfer

Sept 2025 – Dec 2025

- Built an audio style transfer system using direct Mel-spectrogram optimization with TensorFlow, transferring timbral and spectral characteristics of a style audio onto a content audio without training any neural network.
- Designed a combined loss function (content L2 loss + spectral flux and mean/std style loss) weighted $\alpha = 0.05$, $\beta = 0.95$; optimized using the Adam optimizer with tf.GradientTape for direct spectrogram updates, achieving over 90% reduction in total and style loss.
- Quantified style transfer quality via MSE-based similarity analysis (style-to-content ratio of 46.76), and reconstructed audio using Librosa's inverse Mel transformation with Griffin-Lim phase recovery.

Echoes Reborn: A Comprehensive Pipeline for Historic Piano Audio Restoration

Sept 2024 – Jan 2025

- Restored degraded historic piano recordings affected by noise, missing segments, and loss of musical fidelity by developing an end-to-end audio restoration pipeline. The goal is to restore both the spectral content and temporal coherence of the audio using Deep Neural Network techniques.
- Investigated early GAN-based denoising and inpainting approaches, then refined the final method toward a 3-layer CNN classifier, a Convolutional Autoencoder (CAE) denoiser with STFT-based loss, and LPC-based audio reconstruction.
- Implemented audio feature extraction and reconstruction with Librosa on the MusicNet historic piano dataset, supporting both spectral restoration and temporal coherence preservation.
- Improved system performance with 21.21 dB SNR gain in denoising and 99.92% classification accuracy on noisy versus clean audio samples.

Bidirectional Indian Sign Language Translation System

July 2022 – May 2023

- Built a complete two-way ISL translation system: (1) ISL to Text/Speech using CNN-based gesture recognition with preprocessing and feature extraction, deployed as an Android app; (2) Text/Speech to ISL using NLP techniques including tokenization, stop word removal, and lemmatization, also deployed on Android.
- Implemented the ISL-to-Text/Speech phase by building the preprocessing-to-classification pipeline for gesture input, using OpenCV and TensorFlow with an EfficientNet-B0-based model, then deploying it on Android with TensorFlow Lite.
- Developed the Speech/Text-to-ISL phase by combining speech capture, text preprocessing, rule-based sentence restructuring, and sign mapping, while integrating Python-based speech recognition and text-to-speech components to support practical end-to-end communication.
- Published in the Indian Journal of Computer Science and Engineering (IJCSE), 2023. [doi:10.21817/indjcse/2023/v14i3/231403030](https://doi.org/10.21817/indjcse/2023/v14i3/231403030)

TECHNICAL SKILLS

Programming Languages: Python, C/C++, JavaScript, PHP, SQL, Dart, TypeScript

AI,ML, and NLP Tools: NLTK, Tensorflow, Pandas, NumPy, Librosa, Pyaudio, WavGAN, PyTorch, Keras, Sklearn, OpenCV, NMT

Frontend Development: React.js, Next.js, TypeScript, Redux, Axios, HTML/CSS, Tailwind CSS, Ant Design, shadcn/ui, Figma, Responsive Design, Vite

Big Data Tools: Databricks, Hadoop, Apache Spark, Map Reduce, Power BI, HIVE, HBASE, Talend ETL, IBM SPSS

Backend Development: Node.js, Express.js, FastAPI, Flask, REST APIs, GraphQL

Tools & Cloud: Docker, Git, GitHub Actions, AWS (EC2, S3), GCP, Azure